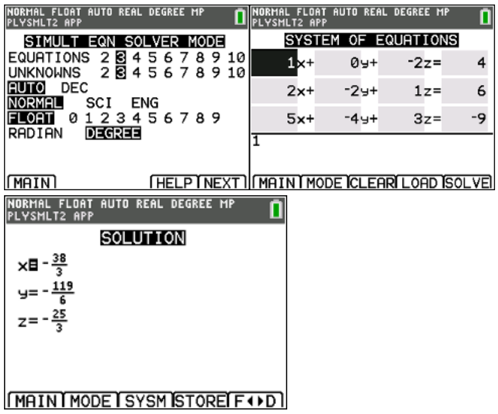
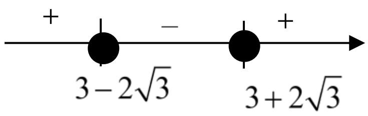
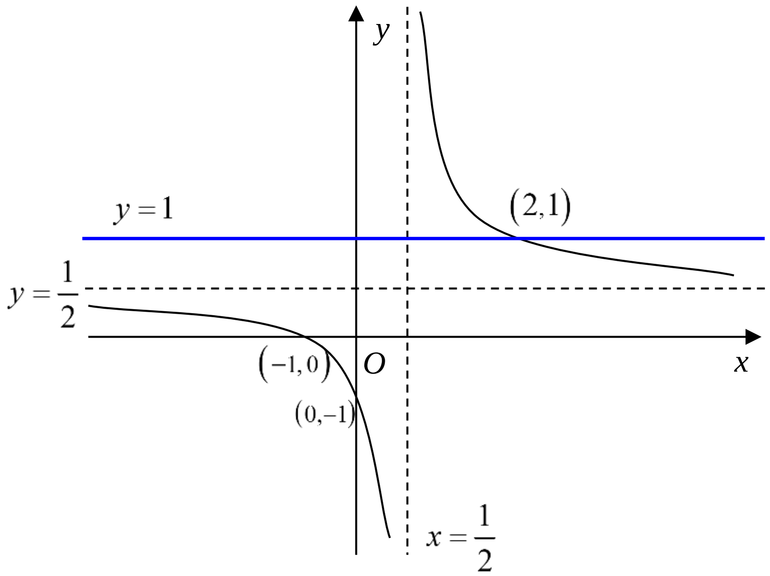
2013 A Level paper 1 Suggested Solutions

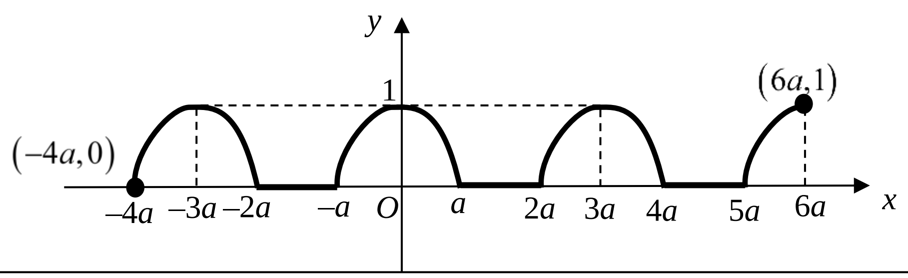
Qn 1	Suggested Solutions
(i)	$p: x-2z=4$ $q: 2x-2y+z=6$ $r: 5-4y+3z=-9$ Using GC, coordinates of the point of intersection $=\left(-\frac{38}{3},-\frac{119}{6},-\frac{25}{3}\right)$ 

Qn 2	Suggested Solutions
	$y=\frac{x^2+x+1}{x-1}$ $(x-1)y=x^2+x+1$ $x^2+(1-y)x+(1+y)=0$ For the equation to have real roots, $(1-y)^2-4(1)(1+y)\geq 0$ $y^2-2y+1-4-4y\geq 0$ $y^2-6y-3\geq 0$ $(y-3)^2-12\geq 0 \quad [\because A^2-B^2=(A+B)(A-B)]$ $\left(y-3-2\sqrt{3}\right)\left(y-3+2\sqrt{3}\right)\geq 0$ Discriminant of the quadratic equation ≥ 0  $\left\{y\in\mathbb{R}: y\leq 3-2\sqrt{3} \quad \text{or} \quad y\geq 3+2\sqrt{3}\right\}$

Qn 3	Suggested Solutions
(i)	$y=\frac{x+1}{2x-1}=\frac{1}{2}+\frac{3}{2(2x-1)}$ Perform Long Division to find the asymptote 
(ii)	$\frac{x+1}{2x-1}<1$ Draw y =1 onto the above curve and find the point of intersection **From the graph,** $x<\frac{1}{2}$ or $x>2$

Qn 4	Suggested Solution
(i)	$w^3=(1+2\text{i})^3$ $=1+3(1)^2(2\text{i})+3(1)(2\text{i})^2+(2\text{i})^3$ $=-11-2\text{i}$ Refer to MF26 for $(x+y)^n$
(ii)	Since w is a root, $a(1+2\text{i})^3+5(1+2\text{i})^2+17(1+2\text{i})+b=0$ $a(-11-2\text{i})+2+54\text{i}+b=0$ $-11a+2+b-2a\text{i}+54\text{i}=0$ comparing the real and imaginary parts, $-2a+54=0$ and $-11a+2+b=0$ $\therefore a=27$ and $b=295$
(iii)	$27z^3+5z^2+17z+295=0$ since the equation has real coefficients, if $w=1+2\text{i}$ is a root, $w^*=1-2\text{i}$ is also a root.

Qn 4	Suggested Solution
	$\left[z-(1+2\text{i})\right]\left[z-(1-2\text{i})\right]=\left[(z-1)-2\text{i}\right]\left[(z-1)+2\text{i}\right]$ $=(z-1)^2-(2\text{i})^2$ $=z^2-2z+5$ $27z^3+5z^2+17z+295=(z^2-2z+5)(pz+q)$ coeff of z^3 : $p=27$ constant: $q=\frac{295}{5}=59$ $\therefore$ the roots are $1+2\text{i},\ 1-2\text{i}$ and $\frac{-59}{27}$. $\boxed{\text{Let } pz+q=0 \text{ to find the last root}}$

Qn 5	Suggested Solutions
(i)	 Note: $y=\sqrt{\left(1-\frac{x^2}{a^2}\right)}$ is the upper portion of the ellipse with centre origin and x-distance a units and y-distance 1 unit. $\because y^2=1-\frac{x^2}{a^2}\Leftrightarrow\frac{x^2}{a^2}+y^2=1$ This graph has a period of 3a units. Label clearly the x and y values of the end points.

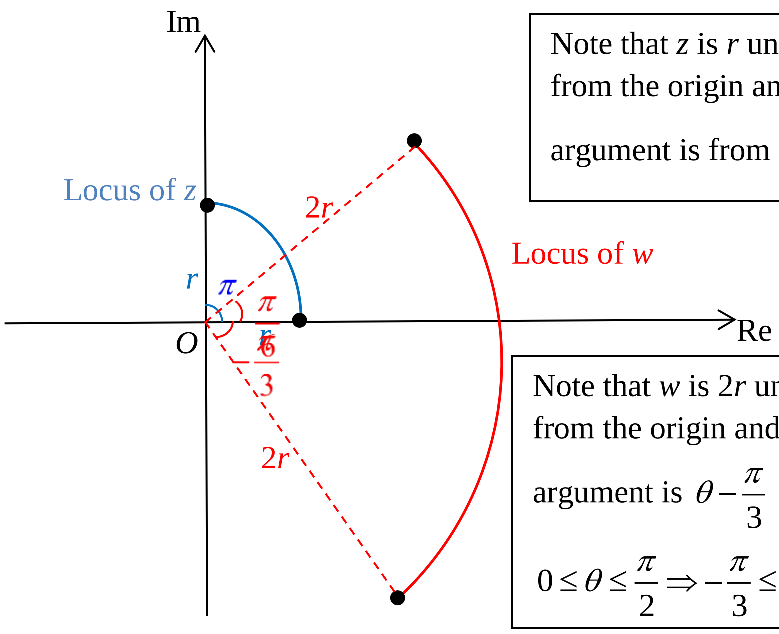
Qn 5	Suggested Solutions
(ii)	$\int_{\frac{1}{2}a}^{\frac{\sqrt{3}}{2}a} \mathrm{f}\left(x\right)\mathrm{d}x$ $= \int_{\frac{1}{2}a}^{\frac{\sqrt{3}}{2}a} \sqrt{\left(1-\frac{x^2}{a^2}\right)}\,\mathrm{d}x$ $= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sqrt{\left(1-\frac{\left(a\sin\theta\right)^2}{a^2}\right)}\,a\cos\theta\mathrm{d}\theta$ $= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sqrt{\left(\cos^2\theta\right)}\,a\cos\theta\ \mathrm{d}\theta$ $= a\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \cos^2\theta\ \mathrm{d}\theta$ $= \frac{a}{2}\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \left(\cos 2\theta+1\right)\mathrm{d}\theta$ $= \frac{a}{2}\left[\frac{1}{2}\sin 2\theta+\theta\right]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$ $= \frac{a}{2}\left[\frac{1}{2}\sin\frac{2\pi}{3}+\frac{\pi}{3}-\left(\frac{1}{2}\sin\frac{\pi}{3}+\frac{\pi}{6}\right)\right]$ $= \frac{a}{2}\left[\frac{\sqrt{3}}{4}+\frac{\pi}{3}-\left(\frac{\sqrt{3}}{4}+\frac{\pi}{6}\right)\right]$ $= \frac{a\pi}{12}$ $x = a\sin\theta$ $\frac{\mathrm{d}x}{\mathrm{d}\theta} = a\cos\theta$ When $x = \frac{1}{2}a$, $\frac{1}{2}a = a\sin\theta \Rightarrow \theta = \frac{\pi}{6}$ When $x = \frac{\sqrt{3}}{2}a$, $\frac{\sqrt{3}}{2}a = a\sin\theta \Rightarrow \theta = \frac{\pi}{3}$ Use Double Angle Formula

Qn 6	Suggested Solutions
(i)	Since O,A,B,C lie on the same plane, $\overrightarrow{OC}$ can be expressed as a sum of a scalar multiple of $\mathbf{a}$ and a scalar multiple of $\mathbf{b}$. **Alternatively,** The equation of plane OAB: $\mathbf{r} = \mathbf{0} + \lambda\mathbf{a} + \mu\mathbf{b}$, $\lambda, \mu \in \mathbb{R}$. Since C lies on plane OAB, $\overrightarrow{OC}$ satisfies the equation of the plane, i.e. $\mathbf{c} = \lambda\mathbf{a} + \mu\mathbf{b}$ for some $\lambda, \mu \in \mathbb{R}$.
(ii)	Using ratio theorem, $\overrightarrow{ON} = \frac{3\mathbf{c}+4\mathbf{a}}{7}$

Qn 6	Suggested Solutions
(iii)	$\overrightarrow{OM}=\frac{1}{2}\mathbf{b}$ Area of ΔONC = Area of ΔOMC $\frac{1}{2}\left\|\overrightarrow{ON}\times\overrightarrow{OC}\right\|=\frac{1}{2}\left\|\overrightarrow{OC}\times\overrightarrow{OM}\right\|$ $\left\|\frac{3\mathbf{c}+4\mathbf{a}}{7}\times\mathbf{c}\right\|=\left\|\frac{1}{2}\mathbf{c}\times\mathbf{b}\right\|$ $\left\|\frac{3}{7}\mathbf{c}\times\mathbf{c}+\frac{4}{7}\mathbf{a}\times\mathbf{c}\right\|=\left\|\frac{1}{2}\mathbf{c}\times\mathbf{b}\right\|$ $\left\|\frac{4}{7}\mathbf{a}\times\mathbf{c}\right\|=\left\|\frac{1}{2}\mathbf{c}\times\mathbf{b}\right\|$ $\left\|\frac{4}{7}\mathbf{a}\times(\lambda\mathbf{a}+\mu\mathbf{b})\right\|=\left\|\frac{1}{2}(\lambda\mathbf{a}+\mu\mathbf{b})\times\mathbf{b}\right\|$ $\left\|\frac{4}{7}\lambda\mathbf{a}\times\mathbf{a}+\frac{4}{7}\mu\mathbf{a}\times\mathbf{b}\right\|=\left\|\frac{1}{2}\lambda\mathbf{a}\times\mathbf{b}+\frac{1}{2}\mu\mathbf{b}\times\mathbf{b}\right\|$ $\left\|\frac{4}{7}\mu\mathbf{a}\times\mathbf{b}\right\|=\left\|\frac{1}{2}\lambda\mathbf{a}\times\mathbf{b}\right\|$ Since $\mu,\lambda>0$, $\frac{4}{7}\mu=\frac{1}{2}\lambda\Rightarrow\lambda=\frac{8}{7}\mu$ Key concepts to know: 1. Area of $\Delta ABC=\frac{1}{2}\left\|\overrightarrow{AB}\times\overrightarrow{AC}\right\|$ 2. $\mathbf{a}\times(\mathbf{c}+\mathbf{b})=\mathbf{a}\times\mathbf{c}+\mathbf{a}\times\mathbf{b}$ 3. $\mathbf{c}\times\mathbf{c}=\mathbf{0}$ 4. $k\mathbf{a}\times\mathbf{c}=-\mathbf{c}\times k\mathbf{a}$

Qn 7	Suggested Solutions
(i)	$p=128\left(\frac{2}{3}\right)^{n-1}$ [Use $U_n=ar^{n-1}$] $\ln p=\ln\left(128\left(\frac{2}{3}\right)^{n-1}\right)$ $\ln p=\ln128+\ln\left(\frac{2}{3}\right)^{n-1}$ $\ln p=7\ln2+(n-1)\ln2-(n-1)\ln3$ $\ln p=(n+6)\ln2+(-n+1)\ln3$ $A=1,B=6,C=-1,D=1$
(ii)	$S_\infty=\frac{a}{1-r}=\frac{128}{1-\left(\frac{2}{3}\right)}=384$ Since $S_n<S_\infty=384$, the total length of string cut off can never be greater than 384 cm

(iii)	$S_n > 380$ $\dfrac{a\left(1-r^n\right)}{1-r} > 380$ $\quad$ [Subt in values of a and r] $384\left(1-\left(\dfrac{2}{3}\right)^n\right) > 380$ $1-\left(\dfrac{2}{3}\right)^n > \dfrac{380}{384}$ $\left(\dfrac{2}{3}\right)^n < \dfrac{1}{96}$ $n\ln\dfrac{2}{3} < \ln\dfrac{1}{96}$ $\quad$ [Switch the sign when you divide by a negative number] $n > 11.257$ Therefore, 12 pieces must be cut off before the total length cut off is greater than 380 cm.

Qn 8	Suggested Solutions
(i)	$w = \left(1-\mathrm{i}\sqrt{3}\right)z$ $= 2\mathrm{e}^{-\frac{\pi}{3}i} r\mathrm{e}^{\mathrm{i}\theta}$ $\quad$ $\left[\because \left\|1-\mathrm{i}\sqrt{3}\right\| = 2, \arg\left(1-\mathrm{i}\sqrt{3}\right) = -\dfrac{\pi}{3}\right]$ $= 2r\mathrm{e}^{\left(\theta-\frac{\pi}{3}\right)i}$ $\|w\| = 2r\,;$ $\quad$ $\arg(w) = \theta - \dfrac{\pi}{3}$
(ii)	 Note that z is r units away from the origin and its argument is from 0 to $\dfrac{\pi}{2}$ Note that w is $2r$ units away from the origin and its argument is $\theta-\dfrac{\pi}{3}$ $0 \le \theta \le \dfrac{\pi}{2} \Rightarrow -\dfrac{\pi}{3} \le \theta - \dfrac{\pi}{3} \le \dfrac{\pi}{6}$

(iii)	
	$$\arg\left(\frac{z^{10}}{w^2}\right)=\pi$$ $$\arg z^{10}-\arg w^2=\pi$$ $$10\arg z-2\arg w=\pi$$ $$10\theta-2\left(\theta-\frac{\pi}{3}\right)=\pi$$ $$8\theta+\frac{2\pi}{3}=\pi$$ $$\theta=\frac{\pi}{24}$$

Qn 9	Suggested Solutions
(ii)	$\mathrm{f}(r)-\mathrm{f}(r-1)=2r^3+3r^2+r+24-\left[2(r-1)^3+3(r-1)^2+(r-1)+24\right]$ [Replace r by r-1] $=2r^3+3r^2+r+24-[2r^3-6r^2+6r-2+3r^2-6r+3+r-1+24]$ $=2r^3+3r^2+r+24-[2r^3-3r^2+r+24]$ $=6r^2$ $\therefore a=6.$ $6r^2=\mathrm{f}(r)-\mathrm{f}(r-1)$ $6\sum_{r=1}^{n}r^2=\sum_{r=1}^{n}\left[\mathrm{f}(r)-\mathrm{f}(r-1)\right]$ $=\left[\begin{array}{l}\mathrm{f}(1)-\mathrm{f}(0)\\+\mathrm{f}(2)-\mathrm{f}(1)\\+\mathrm{f}(3)-\mathrm{f}(2)\\+\quad\vdots\\+\mathrm{f}(n)-\mathrm{f}(n-1)\end{array}\right]$ [Use Method of Differences] $=\mathrm{f}(n)-\mathrm{f}(0)$ $=2n^3+3n^2+n+24-24$ $=n(2n^2+3n+1)$ $\sum_{r=1}^{n}r^2=\frac{n}{6}(n+1)(2n+1).$

(iii)	$\sum_{r=1}^{n} f(r) = \sum_{r=1}^{n} [2r^3 + 3r^2 + r + 24]$ $= \sum_{r=1}^{n} \left(2r^3 + r\right) + \sum_{r=1}^{n} [3r^2 + 24]$ $= \sum_{r=1}^{n} r(2r^2 + 1) + 3\sum_{r=1}^{n} r^2 + \sum_{r=1}^{n} 24$ $= \frac{n(n+1)(n^2+n+1)}{2} + \frac{3n(n+1)(2n+1)}{6} + 24n$ [Use (ii) results] $= \frac{n}{2}\left[(n+1)(n^2+n+1) + (n+1)(2n+1) + 48\right]$

10	**Suggested Solutions**
(i)	$\frac{\mathrm{d}z}{\mathrm{d}x} = 3 - 2z$ $\int \frac{1}{3-2z}\,\mathrm{d}z = \int 1\,\mathrm{d}x$ $-\frac{1}{2}\ln\lvert 3-2z\rvert = x + c$, where c is a constant. [Use $\int \frac{f'(x)}{f(x)}\mathrm{d}x = \ln\lvert f(x)\rvert + C$] Since $z < \frac{3}{2}$ i.e $3 - 2z > 0$, we have $\ln\left(3-2z\right) = -2x - 2c$ $3 - 2z = \mathrm{e}^{-2x-2c}$ $z = \frac{3}{2} - \frac{1}{2}\mathrm{e}^{-2x}\mathrm{e}^{-2c}$ $z = \frac{3}{2} + A\mathrm{e}^{-2x}$ where $A = -\frac{1}{2}\mathrm{e}^{-2c}$ Note: $A < 0$ since $\mathrm{e}^{-2c} > 0$
(ii)	$\frac{\mathrm{d}y}{\mathrm{d}x} = z$ $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3}{2} + A\mathrm{e}^{-2x}$ and $A < 0$ $y = \int \frac{3}{2} + A\mathrm{e}^{-2x}\ \mathrm{d}x$ $y = \frac{3}{2}x - \frac{A}{2}\mathrm{e}^{-2x} + D$, where D is a constant and A<0

(iii)	$\frac{dy}{dx}=\frac{3}{2}+Ae^{-2x}$ --- (1) $\frac{d^2y}{dx^2}=-2Ae^{-2x}$ From (1), we have $Ae^{-2x}=\frac{dy}{dx}-\frac{3}{2}$ Hence, $\frac{d^2y}{dx^2}=-2\left(\frac{dy}{dx}-\frac{3}{2}\right)$ $\frac{d^2y}{dx^2}=-2\frac{dy}{dx}+3$ $\therefore a=-2,\ \ b=3$

Q11	**Suggested Solutions**
(i)	$x=3t^2, \qquad y=2t^3$ $\frac{dx}{dt}=6t \qquad \frac{dy}{dt}=6t^2$ $\Rightarrow \ \frac{dy}{dx}=\frac{6t^2}{6t}=t$ Equation of tangent with parameter t is $y-2t^3=t\left(x-3t^2\right)$ [Use $y - y_1 = m(x - x_1)$] $y=tx-3t^3+2t^3$ $=tx-t^3$
(ii)	Equation of tangent at P is $y=px-p^3$ [Replace t by p] Equation of tangent at Q is $y=qx-q^3$ [Replace t by q] Point of intersection is $px-p^3=qx-q^3$ $px-qx=p^3-q^3$ $x=\frac{p^3-q^3}{p-q}$ $=\frac{(p-q)\left(p^2+pq+q^2\right)}{p-q}$ $=p^2+pq+q^2$(shown) When $x=p^2+pq+q^2$, $y=p\left(p^2+pq+q^2\right)-p^3$ $=p^2q+pq^2$

	Since $pq=-1$, $x=p^2+pq+q^2$ $=(p+q)^2-pq$ $=(p+q)^2+1$ $y=p^2q+pq^2$ $=pq(p+q)$ $=-(p+q)$ $y^2+1=\left[-(p+q)\right]^2+1$ $=(p+q)^2+1$ $=x$ $\therefore R$ lies on the curve $x=y^2+1$ (shown)
(iii)	$x=3t^2, \qquad y=2t^3 ---------(1)$ $x=y^2+1 --------------(2)$ Sub (1) into (2) $3t^2=\left(2t^3\right)^2+1$ $3t^2=4t^6+1$ $4t^6-3t^2+1=0$ (shown) Let $r=t^2$ $4r^3-3r+1=0$ $(r+1)\left(4r^2-4r+1\right)=0$ $(r+1)(2r-1)^2=0$ $r=-1$ or $r=\frac{1}{2}$ $t^2=-1$ $t^2=\frac{1}{2}$ (No real roots) $t=\pm\frac{1}{\sqrt{2}}$ Since $y\geq 0$, $t=\frac{1}{\sqrt{2}}$ [since $y=2t^3$]

	$x = 3\left(\frac{1}{\sqrt{2}}\right)^2 \qquad y = 2\left(\frac{1}{\sqrt{2}}\right)^3$ $x = \frac{3}{2} \qquad y = \frac{1}{\sqrt{2}}$ The coordinates of M is $\left(\frac{3}{2}, \frac{1}{\sqrt{2}}\right)$.
(iv)	Area of shaded region $= \int_0^{\frac{1}{\sqrt{2}}} x_L - x_C \, \mathrm{d}y$ $= \int_0^{\frac{1}{\sqrt{2}}} \left(y^2+1\right)\mathrm{d}y - \int_0^{\frac{1}{\sqrt{2}}} \left(3t^2\right)6t^2 \, \mathrm{d}t$ [Note that $x_C = 3t^2, \frac{dy}{dt} = 6t^2$ and $= \left[\frac{1}{3}y^3 + y\right]_0^{\frac{1}{\sqrt{2}}} - 18\left[\frac{1}{5}t^5\right]_0^{\frac{1}{\sqrt{2}}}$ you must find the corresponding limits for t] $= \left(\frac{1}{6\sqrt{2}} + \frac{1}{\sqrt{2}}\right) - \frac{18}{5}\left(\frac{1}{4\sqrt{2}}\right)$ $= \frac{4}{15\sqrt{2}} = \frac{2\sqrt{2}}{15}$

	Alternatively, For curve L, when $y=0$, $x=1$ Area of shaded region $=\int_0^{\frac{3}{2}} y_C \, \mathrm{d}x - \int_1^{\frac{3}{2}} y_L \, \mathrm{d}x$ $=\int_0^{\frac{1}{\sqrt{2}}} 2t^3 \left(6t\right) \mathrm{d}t - \int_1^{\frac{3}{2}} \sqrt{x-1}\, \mathrm{d}x$ $=12\left[\frac{t^5}{5}\right]_0^{\frac{1}{\sqrt{2}}} - \frac{2}{3}\left[\left(x-1\right)^{\frac{3}{2}}\right]_1^{\frac{3}{2}}$ $=\frac{12}{5}\left(\frac{1}{4\sqrt{2}}\right)-\frac{2}{3}\left(\frac{1}{2}\right)^{\frac{3}{2}}$ $=\frac{3}{5\sqrt{2}}-\frac{1}{3\sqrt{2}}$ $=\frac{4}{15\sqrt{2}}=\frac{2\sqrt{2}}{15}$